

Q1. What is the difference between condensers and capacitors?

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Through e-mail

A1. From electrical engineering point of view, condensers and capacitors are the same. However, you may often come across condensers also being referred to as devices used to condense vapour into liquid, a range of laboratory glassware used to remove heat from fluids, heat exchangers installed in steam-electric power stations to condense turbine exhaust steam into water, air coils used in HVAC refrigeration systems, a group of lenses mounted below the stage of an optical microscope to concentrate light, and condenser microphones that convert sound waves into an electrical signal.

An ideal capacitor is characterised by a single constant value called 'capacitance.' This is defined as the ratio of the amount of charge in each conductor to the potential difference between them. The unit of capacitance is thus coulombs per volt, or farads. Higher capacitance indicates that more charge can be stored at a given voltage. In practice, the insulator allows a small amount of current through, called leakage current. The conductors add a series resistance (specifically called equivalent series resistance), and the insulator has an electric field strength limit resulting in a breakdown voltage.

The properties of capacitors in a circuit may determine the resonant frequency and quality factor of a resonant circuit, power dissipation and operating frequency in a digital logic circuit, energy capacity in a high-power system, and many other important aspects.

Q2. What is the working principle behind touchscreen mobiles? How do they differ from ordinary mobiles? What are their drawbacks?

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A2. Mobile phones may use two types of input devices. In reg-

ular mobile phones, a keypad type of device is used, which is mounted separately from the screen of the cellphone. Whereas in touchscreen cellphones, a touch screen is a cellphone display screen that also acts as an input device. The touchscreens are sensitive to pressure; a user interacts with the mobile applications by touching pictures or words on the screen.

Most mobile phone keyboards are basic in that they use a tactile surface you are accustomed to touching, and underneath is a basic rubber peg (black dot) which travels some depth until it encounters resistance in the form of the actual keyboard surface which is sometimes called a 'bubble board.' This is basically a semi-circle of aluminium shaped in the form of a dome and provides that springing effect of key and feedback on your finger when you press down and the button regains its at-rest shape and normal position.

Touchscreen technologies used in mobile phones include resistive, capacitive and surface-wave based system.

Resistive. The resistive system consists of a normal glass panel that is covered with conductive and resistive metallic layers. These two layers are held apart by spacers, and a scratch-resistant layer is placed on top of the whole setup. An electrical current runs through the two layers while the monitor is operational.

When a user touches the screen, the two layers make contact exactly at that spot. The change in the electrical field is noted and the coordinates of the point of contact are calculated by the processor. Once the coordinates are known, a special driver translates the touch into something that the operating system can understand, much as a computer mouse driver translates the movements of a mouse into a click or a drag. The change in the electrical current is registered as a touch event and sent to the controller for processing.

Resistive touchscreen panels are generally more affordable but offer only 75 per cent clarity and the layer

can be damaged by sharp objects. Resistive touchscreen panels are not affected by outside elements such as dust or water.

Capacitive. In the capacitive system, a layer of an electroconductive material (most often indium tin-oxide) that stores electrical charge is placed on the glass panel of the monitor. When a user touches the monitor with his finger, some of the charge is transferred to the user, so the charge on the capacitive layer decreases. This decrease is measured in circuits located at each corner of the monitor. The computer calculates, from the relative differences in charge at each corner, exactly where the touch event took place and then relays that information to the touchscreen driver software.

One advantage of the capacitive system over the resistive system is that it transmits almost 92 per cent of the light emitted from the monitor, whereas the resistive system transmits only about 75 per cent. This gives the capacitive system a much clearer picture than the resistive system. Also, the capacitive system has a very long life (about 225 million clicks). The bad news is that this touchscreen type cannot be activated by contact with inanimate objects (e.g., the gloves that you might be wearing).

There are mainly two subtypes: one cannot register more than one touch at a time, while the other, called 'MultiTouch' (used in Apple iPhone and iPod) does.

Surface wave. Surface-wave technology uses ultrasonic waves that pass over the touchscreen panel. On the monitor of a surface acoustic wave system, two transducers (one sending and the other receiving) are placed along the 'x' and 'y' axes of the monitor's glass plate. Also placed on the glass are reflectors—these reflect an electrical signal sent from one transducer to the other. The receiving transducer is able to tell whether the wave has been disturbed by a touch event at any instant, and can locate it accordingly. When the panel is touched, a portion of the wave is